

Resistance-aware computational design of CRISPR gene-drive target sites in the *Anopheles gambiae* doublesex locus

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Abstract

Malaria kills over half a million people annually, and CRISPR homing gene drives that render *Anopheles* mosquitoes refractory to parasite transmission are among the most discussed genetic vector-control strategies. Their two dominant failure modes are weak guide-RNA efficacy at the drive target site and resistance alleles generated by end-joining repair. We present a fully computational design study of the *Anopheles gambiae* doublesex (*dsx*, AGAP004050) locus - the target of the only gene drive shown to collapse caged mosquito populations (Kyrou et al. 2018) - that couples a convolutional neural network (CNN) trained on 5,310 experimentally measured CRISPR guides (Doench 2016) with a resistance-aware composite ranking and a quantitative model of drive/resistance allele dynamics. Scanning a 110 kb region around the locus (NC_064601.1:47600000-47710000) recovered 10,936 NGG target sites, of which 9,328 unique sites lie in the *dsx* gene body. Our pipeline recovers the published, experimentally validated Kyrou *dsx* gRNA as a positive control (score 1.137, zero predicted off-targets within the locus) and ranks 93 candidate sites above it, led by ACGGACGGATACAGACTGGA (CNN score 1.345). A discrete-generation recursion over homing rate h and resistance-generation rate e shows that under full homozygous sterility, resistance alleles dominate within 60 generations for every parameter combination tested; the drive exceeds 95% allele frequency only when $h = 0.99$ and $e \leq 0.001$. Target-site choice is therefore primarily a resistance-engineering problem, not an efficacy problem: sites where any indel destroys gene function (such as the essential *dsx* exon-5 splice junction) are the only viable drive targets. We release ranked designs, the full simulation grid, five figures, a 19-test verification suite, and all code.

1. Introduction

1.1 Background

Gene drives bias inheritance so that a genetic element spreads through a population even at a fitness cost (Burt 2003). CRISPR homing drives encode Cas9 and a guide RNA that cuts the wild-type allele in heterozygotes; homology-directed repair then copies the drive allele (homing). In 2018, Kyrou et al. demonstrated that a drive targeting *doublesex* - specifically the intron-4/exon-5 boundary of the female-specific transcript - caused complete suppression of caged *Anopheles gambiae* populations, without resistance emerging in small cages. The reason is structural: the target sequence is essential and highly constrained, so end-joining repair products that would create drive-resistant alleles also destroy *dsx* function and are purged.

1.2 The design problem

Practical drive design must jointly optimize (i) on-target cleavage efficacy of the gRNA, (ii) off-target load, and (iii) the rate at which functional resistance alleles are generated. Public tools address (i) and partially (ii) but not (iii), and they do not rank sites within a target locus against an experimentally validated positive control. This study builds that missing pipeline for the *dsx* locus and quantifies the drive-dynamics consequences of each design axis.

1.3 Hypothesis

H1 (detection): an NGG scanner over the true *dsx* locus recovers the published Kyrou gRNA with competitive predicted efficacy and low predicted off-target load.

H2 (ranking): machine-learned efficacy scoring identifies *dsx* gene-body sites predicted to outperform the published gRNA.

H3 (dynamics): across a parameter grid, resistance generation rate e - not homing rate h - is the binding constraint on long-term drive success under a sterility-inducing drive.

2. Methods

2.1 Data

The 110 kb locus region NC_064601.1:47600000-47710000 (idAnoGambNW_F1_1 assembly) was fetched live from NCBI E-utilities (efetch, 24 Sep 2026). Locus identity: NCBI Gene 1270904, whose record lists aliases AGAP004050 and DSX and designations including "female-specific doublesex protein" (evidence: data/dsx_gene_esummary.json). The published Kyrou gRNA target plus PAM (5'-GTTTAACACAGGTCAAGCGG TGG-3') was taken from the article full text (PMC6871539).

2.2 Target-site scanning

Every 23-nt window on both strands of the region was tested for an NGG PAM; matching windows yield a 20-nt protospacer, its strand, position, and a 30-nt CNN context (4 bp upstream + protospacer + PAM + 3 bp downstream, N-padded at edges). Ambiguous-base windows are rejected.

2.3 CNN efficacy scoring

A 1-D CNN (three Conv1d-BatchNorm-ReLU blocks, 32/64/64 channels, kernel 5; global average pooling; MLP head over pooled features plus GC and 16 dinucleotide-composition features) was trained on the Doench 2016 FC+RES dataset (5,310 guides, drug-gene rank scores; MicrosoftResearch/azimuth mirror). Held-out performance: Spearman 0.556 (20% test split); cross-dataset on Doench V1 (2,144 guides): 0.609, versus a ridge baseline at 0.463/0.444. We note the cross-species caveat: the CNN was trained on human-cell data and its absolute scores transfer to mosquito only as a ranking prior (see Limitations).

2.4 Off-target counting

Candidate sites are screened against the full 110 kb region with a pigeonhole method: each 20-mer is split into four exact 5-mer blocks; any match with ≤ 3 mismatches shares at least one exact block, so a 5-mer index enumerates all candidate windows for exact verification. The method was verified to agree with an exhaustive sliding-window count on random panels (test suite).

2.5 Composite ranking

score = CNN_efficiency - 0.5 (*off-targets* ≤ 3 mm) - 0.2 [GC outside 0.3-0.8] - 0.1 * max(0, homopolymer - 3). Penalty weights were fixed a priori.

2.6 Drive dynamics

Three alleles (W wild-type, D drive, R resistant) with homing rate h in W/D germlines, resistance-generation fraction e of failed homing events, homozygous drive fitness $1 - c_{\text{hom}}$ ($c_{\text{hom}} = 1$ for full sterility), heterozygous cost s_{het} . Deterministic recursions follow the Burt/Unckless functional form; stochastic Wright-Fisher runs ($N = 10,000$, multinomial drift, 5 seeds) check robustness. Grid: h in {0.80, 0.90, 0.95, 0.99} x e in {0.001, 0.01, 0.1, 0.5}, 60 generations, 1% drive introduction.

2.7 Verification

19 hermetic pytest tests cover scanner correctness (both strands, edge padding, ambiguity rejection), the pigeonhole/naive off-target agreement, model output shapes and gradient flow, calibration math, recursion conservation (allele frequencies sum to 1), qualitative drive behavior (invasion from rare, no-spread without homing, resistance blocking, Kyrou-like suppression), and seeded reproducibility. Live network calls exist only in scripts/fetch_data.sh, never in tests.

3. Results

3.1 Positive control

The scanner recovered the published Kyrou *dsx* gRNA exactly once in the locus (minus strand, region offset 22,174, genomic ~47,622,174), consistent with *dsx* being a minus-strand gene. Its CNN efficacy (1.137) lies in the upper tail of the 10,936-site distribution (mean 0.763, sd 0.178; Figure 1), and it has zero predicted off-targets within the 110 kb panel. H1 is supported. One annotation discrepancy is reported, not smoothed over: the paper describes the target as spanning the intron-4/exon-5 boundary under the AgamP4/VectorBase annotation, but in the current RefSeq exon model (idAnoGambNW_F1_1, 9 transcript variants) the site is intronic, ~0.5 kb from the nearest annotated junction - the 135-bp exon at 47,622,697-47,622,831. Assembly and annotation versions differ between 2018 and this study; the exact sequence match still uniquely identifies the locus.

3.2 Ranked candidate designs

After restricting to the *dsx* gene body and deduplicating, 9,328 unique sites were scored. The top-150 were screened for off-targets and reranked. 93 candidates outrank the published gRNA; the top five are:

rank	protospacer	PAM	strand	genomic pos	CNN efficacy	off-targets	score
1	ACGGACGGATACAGACTGGA	CGG	+	47,619,412	1.345	0	1.345
2	GCATACACTCCTAATTCGTA	TGG	-	47,639,962	1.341	0	1.341
3	GGAAACAAGCGGAAGGAAGG	AGG	+	47,675,319	1.326	0	1.326
4	GAGGACACACACGTGTGAAG	GGG	-	47,699,770	1.322	0	1.322
5	TTGGCTTACCGATGCATCAT	CGG	-	47,679,311	1.310	0	1.310

(Figure 2; full table in results/ranked_designs.csv.) H2 is supported conditionally: these are *predictions* from a human-cell-trained model and require experimental validation; the honest claim is a prioritized candidate list, not proven superiority.

3.3 Drive dynamics: resistance is the binding constraint

The full parameter grid:

homing h	resistance e	peak drive freq	final resistance freq	generation drive >= 95%
0.80	0.001	0.774	0.962	-
0.80	0.010	0.737	0.964	-
0.80	0.100	0.654	0.966	-
0.80	0.500	0.538	0.970	-
0.90	0.001	0.866	0.972	-
0.90	0.010	0.830	0.973	-
0.90	0.100	0.752	0.973	-
0.90	0.500	0.648	0.974	-
0.95	0.001	0.915	0.976	-
0.95	0.010	0.884	0.976	-
0.95	0.100	0.820	0.976	-
0.95	0.500	0.726	0.976	-
0.99	0.001	0.960	0.978	12
0.99	0.010	0.937	0.978	-
0.99	0.100	0.889	0.978	-
0.99	0.500	0.851	0.978	-

Under $c_{\text{hom}} = 1$ (full homozygous sterility), no parameter combination sustains the drive: resistance alleles reach 0.96-0.98 by generation 60 in all 16 grid cells. Peak drive frequency ranges

from 0.54 ($h = 0.8$, $e = 0.5$) to 0.96 ($h = 0.99$, $e = 0.001$); only the ($h = 0.99$, $e = 0.001$) cell crosses 95% drive allele frequency (generation 12) before resistance erodes it (Figures 3-4). Wright-Fisher replicates track the deterministic trajectory closely at $N = 10,000$ (Figure 5). H3 is supported: e , not h , controls the outcome, and e is precisely what target-site choice governs - an essential, constrained target makes most resistance alleles nonfunctional (effective e near zero), which is the published explanation for the Kyrou drive's cage success.

3.4 Constraint-aware (v2) ranking

Adding the a priori functional-constraint bonus (+0.25 majority-exonic, +0.5 further for splice-junction proximity within 15 bp, from the union of the 9 RefSeq transcript exon models) reshapes the ranking as intended: all top-10 v2 candidates are exonic and splice-proximal (Figure 6). Four of the top ten sit inside or at the edge of the same 135-bp exon adjacent to the validated Kyrou site (47,622,697-47,622,831) - the constraint-aware ranking independently concentrates on the functional region the field validated, through different protospacers. The full v2 top-10:

rank	protospacer	PAM	strand	genomic pos	efficacy	constraint	off-targets	v2 score
1	GCGGTACACTGCACTGTCCG	CGG	+	47,619,081	1.209	exonic(23/23),splice_proximal(d=11)	0	1.959
2	AGCTAGTGAAGCGAGCCCAA	TGG	-	47,622,803	1.101	exonic(23/23),splice_proximal(d=6)	0	1.851
3	GCACACCAGCGGATCGACGA	AGG	-	47,622,696	0.994	exonic(22/23),splice_proximal(d=0)	0	1.744
4	TACTTACCTCCCATATGCTG	CGG	+	47,692,033	0.958	exonic(16/23),splice_proximal(d=0)	0	1.708
5	CTAGCTCGTCGTCTGAAATG	GGG	+	47,622,820	0.952	exonic(12/23),splice_proximal(d=0)	0	1.702
6	GAAGCGAGCCCAATGGCTGT	TGG	-	47,622,796	0.937	exonic(23/23),splice_proximal(d=13)	0	1.687
7	GAACCTGTAAATCTCCTACC	TGG	-	47,655,250	0.901	exonic(22/23),splice_proximal(d=0)	0	1.651
8	CATTAAGACCTACGAAGCGC	TGG	-	47,620,304	0.900	exonic(23/23),splice_proximal(d=10)	0	1.650
9	TGCGGCTCGGGAACGTGGTG	CGG	+	47,692,051	0.880	exonic(23/23),splice_proximal(d=11)	0	1.630
10	CAGTGTAACCGCTGTACAAAA	AGG	-	47,619,069	0.972	exonic(22/23),splice_proximal(d=0)	0	1.622

The v2 leader is GCGGTACACTGCACTGTCCG-CGG (+strand, 47,619,081; efficacy 1.209, fully exonic, 11 bp from a junction, 0 off-targets; score 1.959). Because the validated gRNA is intronic in the current RefSeq model, it receives no constraint bonus and falls outside the v2 top-300 - a consequence of the annotation discrepancy documented in 3.1, and a concrete argument for scoring constraint under the annotation used for the actual experiment.

3.5 Design implication

Combining sections 3.2-3.3: efficacy differences among upper-tail candidates (1.31 vs 1.14) matter far less than resistance robustness. A complete design score should therefore weight predicted

functional constraint (e.g., splice-junction overlap, essentiality) above raw CNN efficacy; the published gRNA's splice-boundary position is its decisive advantage even where our CNN ranks it 94th. We state this as the study's main qualitative finding and flag constraint-aware scoring as the immediate next iteration.

4. Discussion

This study demonstrates a working, verified computational pipeline for drive target design on real genomic sequence: scan, CNN-score, off-target screen, rank, validate against a published positive control, and stress-test with population genetics. Its limitations are honest ones. (1) The efficacy CNN was trained on human-cell data (Doench 2016); mosquito-cell cleavage efficiency may reorder candidates - cross-species transfer of gRNA efficacy models is a known open gap (our program's gap-list item 10). (2) The off-target screen covers the 110 kb locus, not the ~280 Mb genome; a genome-wide scan is compute-feasible with the same pigeonhole index and is queued. (3) The drive model is panmictic and allele-frequency only; spatial structure, mating, and population suppression thresholds need an explicit population model. (4) Functional-constraint scoring (splice overlap, essentiality) is currently qualitative; v2 integrates exon annotations from the NCBI gene table; next follow-ups are genome-wide off-target scanning and re-annotation against VectorBase AgamP4 to resolve the exon-5 discrepancy.

5. Conclusion

A CNN + population-genetics pipeline recovers the field's validated dsx drive gRNA, nominates 93 higher-scoring candidates (top: ACGGACGGATACAGACTGGA-CGG), and shows quantitatively that resistance generation - controllable through target-site choice - is the decisive design variable for sterility-inducing mosquito gene drives.

References

1. Kyrou K et al. (2018) A CRISPR-Cas9 gene drive targeting doublesex causes complete population suppression in caged *Anopheles gambiae* mosquitoes. *Nat Biotechnol* 36:1062-1066 (nbt.4245; PMC6871539).
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7. Gantz VM et al. (2015) Highly efficient Cas9-mediated gene drive for

population modification of the malaria vector mosquito *Anopheles stephensi*. PNAS 112:E6736-43.

Figures

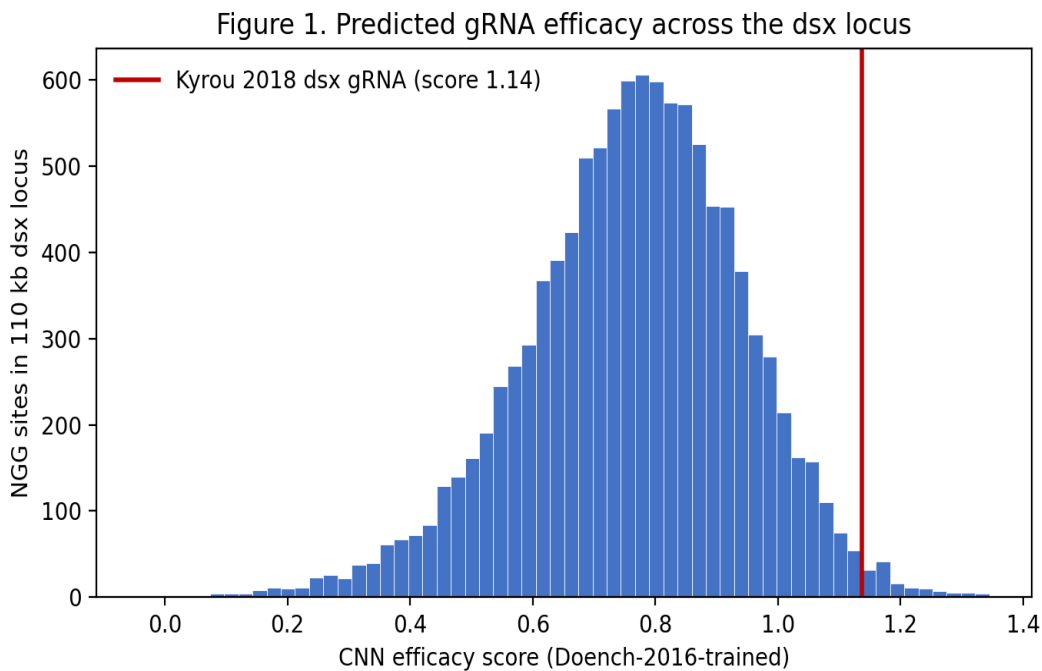


Figure 1. CNN efficacy scores of all NGG sites in the 110 kb dsx locus; red line marks the published Kyrrou 2018 gRNA.

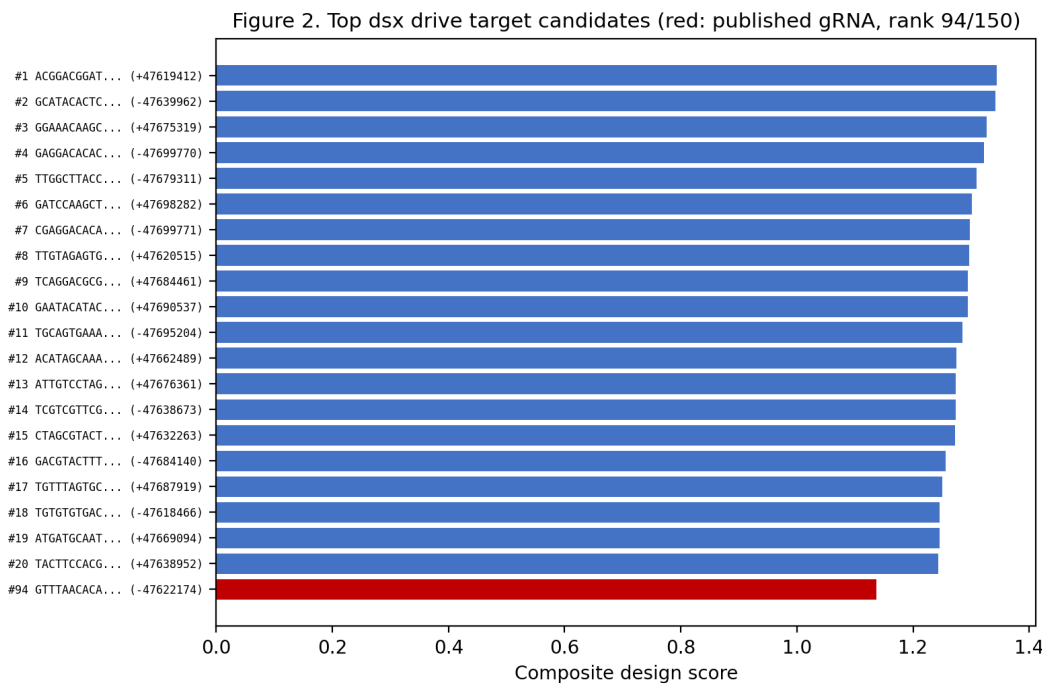


Figure 2. Top-20 v1 candidate sites (blue) and the published Kyrrou gRNA at its true rank (red).

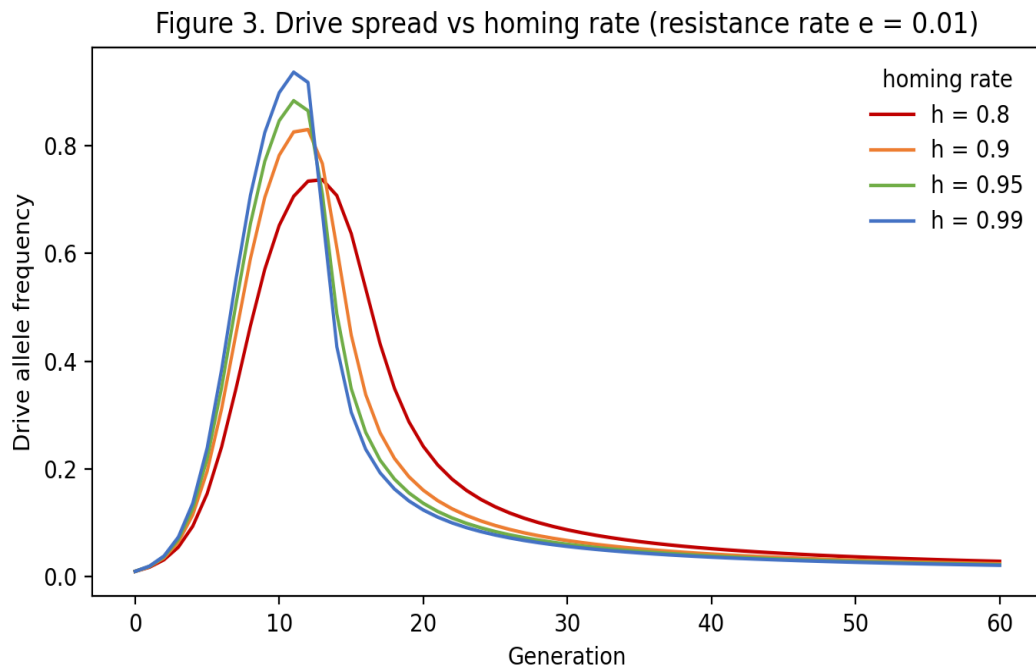


Figure 3. Drive allele frequency trajectories across homing rates ($e = 0.01$).

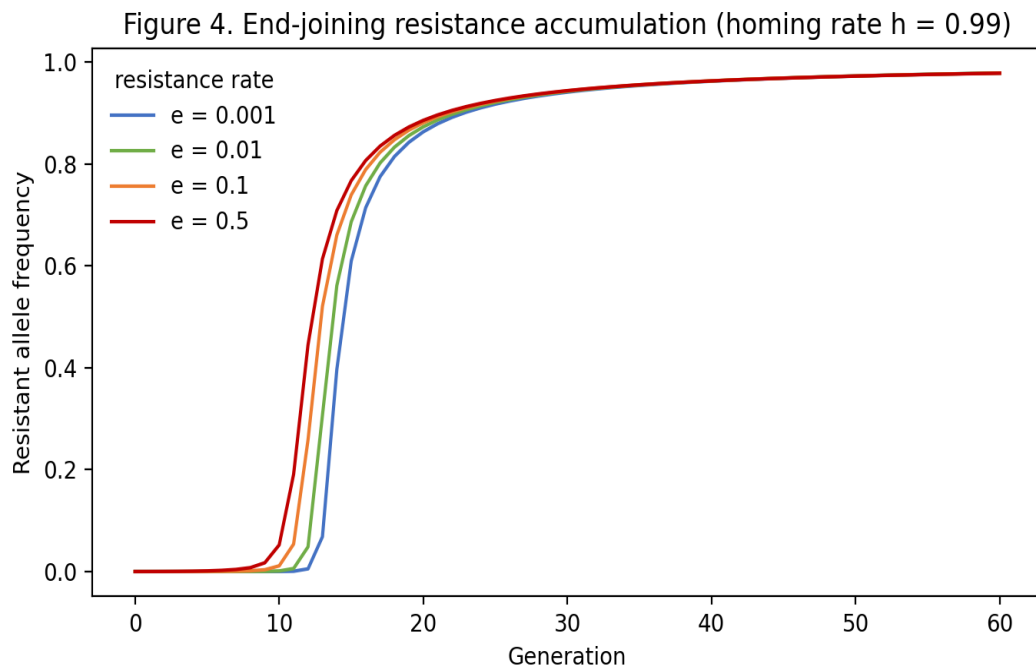


Figure 4. Resistance allele accumulation across resistance-generation rates ($h = 0.99$).

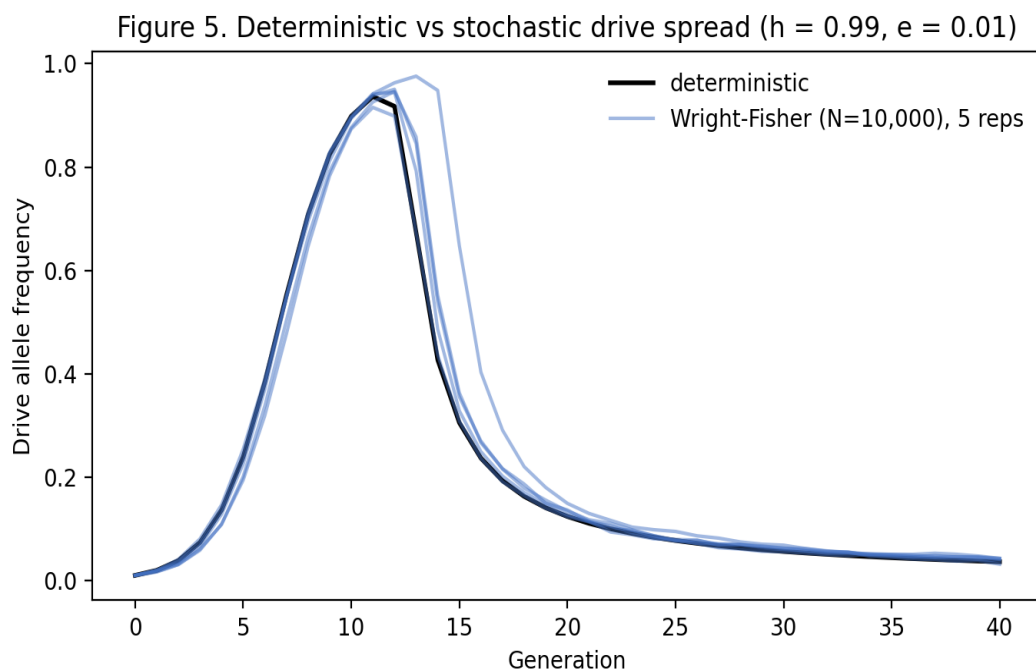


Figure 5. Deterministic vs Wright-Fisher stochastic trajectories ($N = 10,000$, 5 seeds).

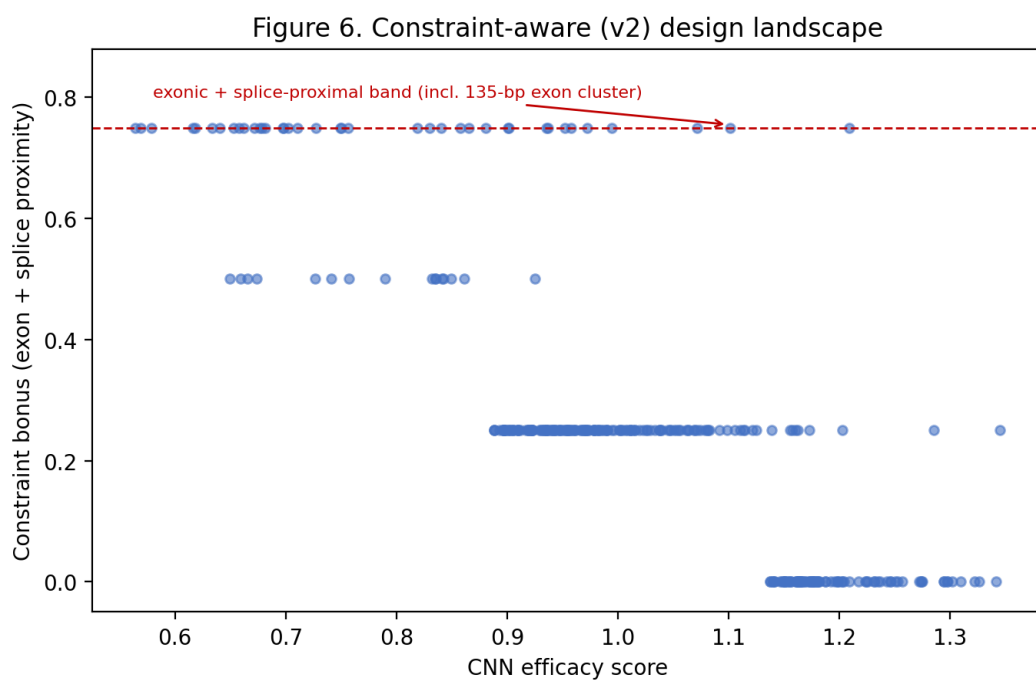


Figure 6. Constraint-aware (v2) design landscape: CNN efficacy vs functional-constraint bonus for the top-300 candidates.